AP PHYSICS 1

Unit 1 • Worksheet 1.2A

Worksheet 1.2A • Displacement, Velocity, and Acceleration

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work, start from a physics principle or equation, and justify your answers. A calculator and the AP Physics 1 equation sheet are allowed. Use $g = 10\,\text{m/s}^2$ and ignore air resistance unless stated.

Question 1

A runner covers 100 m in a straight line in 12.5 s. What is her average speed?

(A) $12.5\,\text{m/s}$

(B) $80\,\text{m/s}$

(C) $0.125\,\text{m/s}$

(D) $8\,\text{m/s}$

Question 2

A car speeds up from $10\,\text{m/s}$ to $25\,\text{m/s}$ in 5 s. What is its average acceleration?

(A) $2\,\text{m/s}^2$

(B) $5\,\text{m/s}^2$

(C) $3\,\text{m/s}^2$

(D) $7\,\text{m/s}^2$

Question 3

An object has velocity $v = -6\,\text{m/s}$ and acceleration $a = +2\,\text{m/s}^2$. What is happening?

(A) It is speeding up in the negative direction.

(B) It is slowing down.

(C) It is at rest.

(D) It is speeding up in the positive direction.

Question 4

An object starts from rest with a constant acceleration of $3\,\text{m/s}^2$. What is its speed after 4 s?

(A) $7\,\text{m/s}$

(B) $12\,\text{m/s}$

(C) $24\,\text{m/s}$

(D) $0.75\,\text{m/s}$

Question 5

For the object in the previous question, how far does it travel in those 4 s?

(A) 12 m
(B) 48 m
(C) 24 m
(D) 6 m

Question 6

A car moving at 30 m/s brakes with an acceleration of magnitude $6\,\text{m/s}^2$. How far does it travel before stopping?

(A) 75 m
(B) 150 m
(C) 5 m
(D) 90 m

Question 7

A ball is thrown straight up at 15 m/s. How long does it take to reach its highest point?

(A) 3 s
(B) 0.67 s
(C) 1.5 s
(D) 15 s

Question 8

At the highest point of its path, a ball thrown straight up has

(A) $v = 0$ and $a = 0$
(B) $v = 0$ and $a = 10\,\text{m/s}^2$ downward
(C) $v = 10\,\text{m/s}$ upward and $a = 0$
(D) $v = 0$ and $a = 10\,\text{m/s}^2$ upward

Question 9 (Free response)

A train starts from rest and accelerates at $0.50\,\text{m/s}^2$ for 40 s. It then brakes with an acceleration of magnitude $1.0\,\text{m/s}^2$ until it stops.

(a) Find its speed at $t = 40$ s.
(b) Find the distance traveled during the first 40 s.
(c) Find the braking distance and the braking time.